Slow network: checks you can do before raising a ticket

Article KB0010026 | Owner: Network Support | Subcategory: Slow network | Applies to: Windows 11 laptops (Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840), macOS 14/15 MacBook Pro/Air, Mobile devices connected to ACME-Corp Wi-Fi, Any device using the Acme Global wired Ethernet ports

Summary

This guide helps users diagnose slow network performance on Acme Global premises before opening a ticket. It walks through quick self-checks such as verifying Wi-Fi band, signal strength, wired connectivity, and site-wide or ISP issues, then outlines the steps service-desk agents should follow to isolate and resolve the problem.

Symptoms

- Internet feels very slow or web pages load slowly
- Video-call applications (Teams, Zoom) show choppy audio/video or frequent freezing
- File downloads or uploads transfer at a fraction of expected speed
- Network icon in system tray shows limited connectivity or a warning triangle
- Ping to internal resources shows latency >100?ms or packet loss

Common causes

- Access point overload - too many clients associated with a single AP, especially on the 2.4?GHz band
- Client device stuck on 2.4?GHz band while 5?GHz offers higher throughput
- Local RF interference (microwaves, cordless phones, neighboring networks) degrading signal quality
- ISP-wide outage or throttling at the building's internet breakout
- Building-wide network congestion (e.g., uplink saturation, misconfigured QoS)
- VPN client routing traffic through a congested tunnel or split-tunnel misconfiguration

Quick self-checks

Before contacting the Service Desk, perform a few basic checks that often reveal the cause of slowness. These steps take less than five minutes and require no special tools. If any check returns abnormal results, note the observation - it will be useful when you speak with an agent.

1. 1. Disconnect from the VPN (GlobalProtect) and test again - some slowness is caused by a busy VPN tunnel.
2. 2. Open a command prompt (Windows) or Terminal (macOS) and run `ping 8.8.8.8`. Note the average round-trip time and any packet loss.
3. 3. Run a speed test (e.g., fast.com or speedtest.net) and record download/upload speeds. Compare to the expected office bandwidth (typically >=100?Mbps down).
4. 4. Look at the Wi-Fi icon: note whether you are connected to "ACME-Corp" and whether the signal bars are full, medium, or low.

Check Wi-Fi band and signal strength

Many laptops automatically prefer the 2.4?GHz band when signal strength drops, even though that band offers lower maximum throughput and is more prone to interference. Verifying which band you are on and moving closer to an access point can immediately improve performance.

1. 1. On Windows, click the network icon -> Properties of the connected SSID -> look for "Band" or "Network band". On macOS, hold Option and click the Wi-Fi menu; the band appears next to the channel number.
2. 2. If you see 2.4?GHz, try to force a switch to 5?GHz: disable Wi-Fi for 10?seconds, then re-enable, or move nearer to an AP and reconnect.
3. 3. After switching, repeat the ping and speed-test steps from the previous section. A noticeable improvement indicates band-related congestion.

Test wired Ethernet (if available)

If your workstation has a docking station or direct Ethernet jack, a wired test eliminates Wi-Fi variables. A good wired connection should show low latency and speeds close to the contracted circuit.

1. 1. Plug an Ethernet cable into the wall jack or docking station and ensure the link LED shows activity.

2. 2. Disable Wi-Fi to force the system to use the wired adapter.
3. 3. Run the same ping and speed-test commands. Compare results to the wireless tests.

Confirm site-wide or ISP problems

Sometimes the slowness affects multiple users or an entire floor, pointing to a shared resource such as the upstream internet link, a core switch, or an ISP issue. Checking with colleagues and looking for service alerts can save time.

1. 1. Ask a nearby colleague if they are experiencing similar slowness (e.g., video call lag, slow file transfers).
2. 2. Visit the internal status page https://status.acme.example or check the Teams channel "IT Help" for any posted network incidents.
3. 3. If no internal alert exists, open a browser and try to reach an external site (e.g., https://www.microsoft.com). If the site fails to load or times out, the problem may be ISP-related.

Service-desk agent troubleshooting

When a user raises a ticket after performing the self-checks, the Service Desk follows a structured flow to isolate the cause. Agents should gather the user's observations, verify layer-1/2 status, and then move to layer-3 checks. The goal is to determine whether the issue is localized to the device, the access point, the floor, or the ISP.

1. 1. **Collect symptom details** - ask for exact timestamps, affected applications, ping/trace results, and speed-test numbers captured by the user.
2. 2. **Verify device connectivity** - confirm the device is associated with the correct SSID, note the band, signal strength (RSSI), and link speed reported by the NIC.
3. 3. **Check AP health** - using the wireless controller, look at client count per AP, channel utilization, and recent error logs for the AP serving the user's location.
4. 4. **Conduct a wired test** - if the user has a dock, request they plug in and retest; compare wireless vs wired results.
5. 5. **Assess floor-wide impact** - query neighboring APs for similar utilization spikes; if multiple users report issues, suspect a congested uplink or core switch.
6. 6. **ISP verification** - if all internal metrics look normal, perform a traceroute to an external host and check for packet loss or high latency beyond the corporate edge; open a provider ticket if needed.
7. 7. **Apply remediation** - based on findings: steer clients to less-used APs, enable band-steering, adjust RF power, or coordinate with the ISP for circuit issues.

Verification and ticket closure

After any corrective action, the agent must verify that the user's experience has returned to acceptable levels before closing the ticket. Document the before/after metrics and the resolution steps taken.

1. 1. Have the user repeat the ping and speed-test tests and confirm values are within baseline (>=50?Mbps download, <30?ms latency to 8.8.8.8, <1?% packet loss).
2. 2. Ask the user to resume their typical workload (e.g., join a Teams call, download a large file) and report any remaining issues.
3. 3. If satisfied, close the ticket with a summary of the root cause and the fix applied; otherwise, escalate per the escalation guidance below.

When to escalate

If the user's self-checks and the Service Desk's initial steps do not identify the cause, or if the problem affects more than ten users on the same floor, escalate to the Network Support group. Include in the escalation: timestamp of onset, affected SSID and band, AP name/controller ID if known, ping/trace results showing loss or latency >100?ms to internal gateways, speed-test comparisons, and any relevant logs from the wireless controller or edge router.

Keywords: slow network, internet slow, Wi-Fi slow, video call lag, file download slow, network latency, high ping, band steering, 2.4 GHz, 5 GHz, access point congestion, ISP outage, GlobalProtect VPN, wired test, speed test, ping 8.8.8.8